

FlippedRAG: Black-Box Opinion Manipulation Adversarial Attacks to Retrieval-Augmented Generation Models

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Abstract

Retrieval-Augmented Generation (RAG) enriches LLMs by dynamically retrieving external knowledge, reducing hallucinations and satisfying real-time information needs. While existing research mainly targets RAG's performance and efficiency, emerging studies highlight critical security concerns. Yet, current adversarial approaches remain limited, mostly addressing white-box scenarios or heuristic black-box attacks without fully investigating vulnerabilities in the retrieval phase. Additionally, prior works mainly focus on factoid Q&A tasks, their attacks lack complexity and can be easily corrected by advanced LLMs. In this paper, we investigate a more realistic and critical threat scenario: adversarial attacks intended for opinion manipulation against black-box RAG models, particularly on controversial topics. Specifically, we propose FlippedRAG, a transfer-based adversarial attack against black-box RAG-like systems. We first demonstrate that the underlying retriever of a black-box RAG can be reverse-engineered and approximated by enumerating critical queries, candidates, and answers, enabling us to train a surrogate retriever. Leveraging the surrogate retriever, we further craft target poisoning triggers, altering very few documents to effectively manipulate both retrieval and subsequent generation, transferring the attack to the original black-box RAG model. Extensive empirical results show that FlippedRAG substantially outperforms baseline methods, improving the average attack success rate by 16.7%. Across four diverse domains, FlippedRAG achieves on average a 50% directional shift in the opinion polarity of

RAG-generated responses, ultimately causing a notable 20% shift in user cognition. Furthermore, we actively evaluate the performance of several potential defensive measures, concluding that existing mitigation strategies remain insufficient against such sophisticated manipulation attacks. These results highlight an urgent need for developing innovative defensive solutions to ensure the security and trustworthiness of RAG systems.

CCS Concepts

• **Security and privacy** → **Domain-specific security and privacy architectures**; • **Information systems** → *Information extraction*; • **Computing methodologies** → *Natural language processing*.

Keywords

RAG; Black-box Adversarial Attack; Opinion Manipulation

ACM Reference Format:

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1 Introduction

Retrieval-Augmented Generation (RAG) combines information retrieval with the generative capabilities of LLMs, enhancing the timeliness of knowledge acquisition and effectively mitigating the hallucination problem of these models [12, 37]. As the application scope of RAG expands, concerns about its security are also increasing. The basic RAG process typically consists of three components: a knowledge base (corpus collection), a retriever, and an LLM. Vulnerabilities arise when some of the retrieved documents are compromised, or prompt are tampered by malicious attackers, a scenario referred to as a manipulation attack against RAG. Previous

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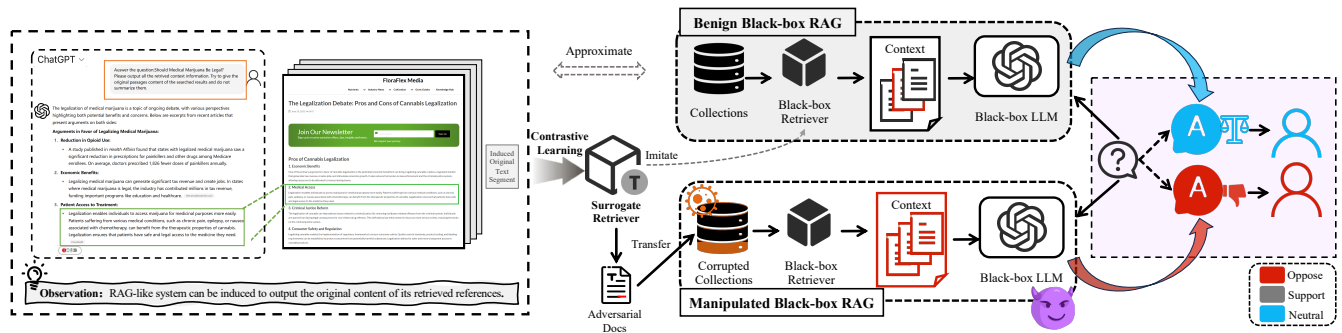


Figure 1: As demonstrated by an example from our pilot study (see left box), we observed that ChatGPT can be induced through user prompts to fully replicate the context segments retrieved in its responses. Motivated by this observation, we can imitate and transparentize the retriever of the black-box RAG model by enumerating critical queries and candidates. The right part illustrates the opinion manipulation attack against black-box RAG-like systems. When queries on controversial topics are submitted to a benign black-box RAG, it typically retrieves documents reflecting diverse viewpoints and generates neutral, objective responses. Conversely, by subtly modifying a minimal number of documents using triggers generated via a surrogate retrieval model, which imitates the black-box retriever, the manipulated RAG exclusively retrieves documents endorsing a specific opinion. This constrains the LLM to generate biased responses and influences users' opinions.

studies have explored various forms of adversarial manipulation attacks, such as adversarial attack on the retriever [19, 22, 30], prompt injection attack [2, 15, 20], jailbreak attack for LLM [8, 17, 38], and poisoning attack targeting the retrieval corpus in RAG [33, 42].

However, previous studies [7, 33, 42] have primarily focused on attacks against RAG systems, which lack feasibility and have fundamental limitations. They mostly address white-box scenario or employ heuristic-based black-box attacks, without thoroughly analyzing the vulnerabilities within the retrieval stage. Furthermore, earlier research has mainly concentrated on factoid question answering tasks, such as questions like “Who is the CEO of OpenAI?” Those factual queries are less impactful, as RAG systems combined with fact-checking and value alignment mechanisms in LLMs are capable of rectifying these errors [29, 31].

Differently, the scenario explored in this paper primarily addresses controversial and opinion-based inquiries in RAG, such as “Should abortion be legal?” These questions are open-ended and require advanced levels of logical reasoning and summarization capabilities from LLMs. Current research on controversial topics is limited. In scenarios involving controversial topics, opinion manipulation can create “Information Bubbles”, which subsequently lead to the homogenization of user opinions [34]. In such cases, individuals' views become easily swayed by the stance of the information they are exposed to. Attacks that manipulate opinions on such questions could potentially inflict deeper harm [10]. Through carefully crafted input, attackers can influence the orientation of content generated by RAG models, thus jeopardizing users' cognitive processes and decision-making abilities.

This paper primarily investigates adversarial opinion manipulation targeting the retriever in black-box RAG-like systems, which aligns with realistic and practical scenarios. The threat model presented here can be characterized as follows: the adversary can only query the RAG-like system and does not have access to the complete knowledge base or corpus, the retriever, or the parameters of the

RAG. The attacker is only capable of injecting limited adversarially modified candidate texts into the corpus, while the retriever and the LLM remain black-boxed, intact, and unmodifiable.

To address aforementioned challenges, in this paper, we propose FlippedRAG, a black-box attack method, to explore the reliability of RAG in controversial topics and investigate its impact on user cognition, as shown in Figure 1. By employing black-box retriever imitation [19] for opinion manipulation, we reveal specific vulnerabilities in RAG-like systems. Our motivation is based on the observation in Figure 1. **When we queried ChatGPT (as of March 2025, with web search enabled, functioning as a typical RAG-like application) on a specific topic and explicitly instructed it to output the original content of its retrieved references, it faithfully reproduced the exact text segments from the retrieved candidates. The reproduced outputs precisely matched the corresponding content on the web pages linked in the reference citations and reflect the functionality of relevance matching.** It allow us to designed a retriever imitation method for black-box RAG to address the limitations of white-box attacks or heuristic-based attacks against such systems. Specifically, we utilize carefully designed instructions to obtain the retrieval results within the RAG model. Subsequently, based on the constructed (query, candidate) pairs from retrieval list, we train a surrogate model using the extracted retrieval data [19, 30] to approximate and transparentize the relevance preferences of the retriever in the black-box RAG model.

Based on this surrogate model, we develop an attack strategy aimed at manipulating the opinions of candidate documents. By attacking this surrogate model, we generate adversarial opinion manipulation triggers, which are then transferred to the target RAG corpus, as shown in the right part of Figure 1. We conduct experiments on opinion datasets encompassing multiple topics to validate the effectiveness and scope of the attack strategy without relying on internal knowledge of the RAG model. The experimental results

Table 1: Comparison of existing RAG attacks. DoS denotes denial-of-service and "-" denotes not applicable.

Methods	Query type	Scenario	LLM Capability Utilization	Depth
PoisonedRAG [42]	Closed-ended, Factoid	White-box/Black-box	Logical reasoning	–
BadRAG [33]	Open-ended, Factoid, DoS	White-box	Logical reasoning	Sentiment
Phantom [4]	Open-ended, Factoid, DoS	White-box	Instruction-following	Sentiment
GARAG [7]	Closed-ended, Factoid	Gray-box	Noise resistance	–
FlippedRAG	Controversial, Open-ended	Black-box	Induction, Logical reasoning	Perspectives

reveal that the proposed attack strategy can significantly alter the opinion polarity of the content generated by RAG. Furthermore, we discuss the efficacy of potential defense mechanisms and conclude that they are insufficient in mitigating this type of attack. This not only highlights the vulnerability of the retrieval model and RAG model but, more importantly, underscores the potential negative impact on user cognition and decision-making. Such vulnerabilities can increase the risk of misleading users into accepting incorrect or biased information.

Our major contributions are as follows:

(1) We propose FlippedRAG, a novel transfer-based opinion manipulation attack against black-box RAG-like systems. It realistically reflects practical threat scenarios, and to our knowledge, we are among the first to study adversarial opinion manipulation targeting open-ended and controversial topics in RAG-based models.

(2) We propose a new approach to construct high-quality pseudo-relevant contrastive pairs for training surrogate retrieval models in black-box RAG. It highlights risks regarding context data leakage in LLM applications. Extensive automated evaluations combined with user studies confirm that our method effectively manipulates the opinions generated by RAG models, ultimately influencing user perception toward specifically targeted stance.

(3) We thoroughly evaluate existing defense measures and find that they fail to effectively prevent or reduce the impact of FlippedRAG, underscoring the urgent need for novel and more robust mitigation strategies.

2 Related Works

2.1 Adversarial Retrieval Attack Strategy

The retrieval model is a core component in RAG. Many existing research evaluates its robustness and security by inducing faulty behaviors through adversarial attacks. The adversarial retrieval attack strategy starts with manipulation at the word level. Under white-box setting, Ebrahimi et al. [9] utilize an atomic flip operation, which swaps one token for another, to generate adversarial text, named HotFlip. HotFlip eliminates reliance on rules, but the adversarial trigger it generates usually has incomplete semantics and insufficient grammar fluency. While it can deceive the target model, it cannot evade perplexity-based defenses. Wu et al. [30] also proposed a word substitution ranking attack method called PRADA. To enhance the readability and effectiveness of the trigger, Song et al. [28] propose an adversarial method under white-box setting, named Collision, which uses gradient optimization and beam search to produce the adversarial trigger. Collision method further imposes a soft constraint on collision generation by integrating a language model, reducing the perplexity of the collision.

Furthermore, Liu et al. [19] propose the Pairwise Anchor-based Trigger (PAT) generation under black-box setting. By adding the fluency and semantic consistency constraints, it generates adversarial triggers by optimizing the pairwise loss of top candidates and target candidates with triggers.

2.2 Reliability of RAG

Some studies have investigated the adversarial poisoning attacks on RAG models, including PoisonedRAG[42], BadRAG[33], Phantom[4], and GARAG[7]. The distinctive characteristics of these attack methodologies are summarized in the Table 1. "LLM Capability Utilization" refers to the motivation of leveraging LLMs' capabilities to induce malicious outputs. "Depth" denotes the cognitive penetration depth of RAG attacks. Although both BadRAG[33] and Phantom[4] target user perception, their manipulations remain confined to sentiment level without achieving deeper ideological persuasion.

As RAG is designed to overcome the hallucination problem in LLMs and enhance their generative capabilities, the reliability of content generated by RAG also becomes a major concern. Zhang et al. [36] sought to identify the weaknesses of RAG by analyzing critical components in order to facilitate the injection of the attack sequence and crafting the malicious documents with a gradient-guided token mutation technique. Xiang et al. [31] proposed an isolate-then-aggregate strategy, named RobustRAG, which first obtains responses from each passage of the LLM in isolation and then securely aggregates these isolated responses, to construct the first defense framework against retrieval corruption attacks. These studies are based on white-box scenarios and primarily focus on the robustness of RAG against corrupted and toxic content.

In this paper, we propose the use of adversarial retrieval attack to manipulate the retrieval process in black-box RAG, ensuring that corrupted documents with a specific opinion are ranked as highly as possible, thereby inducing the LLM responses to align with that opinion on controversial topics.

3 Methodology

3.1 Overview

Our objective is to manipulate the opinions expressed in the responses generated by black-box RAG models on controversial topics. We mainly focus on the retrieval component, where manipulated ranking outcomes propagate to bias the LLM's output generation. Zhang et al. [36] attempted to poison context documents to mislead the LLM into generating incorrect content. However, this approach necessitates extensive internal details of the LLM application, rendering it less feasible in real-world scenarios. In the black-box RAG context, the attacker lacks access to internal information of the RAG,

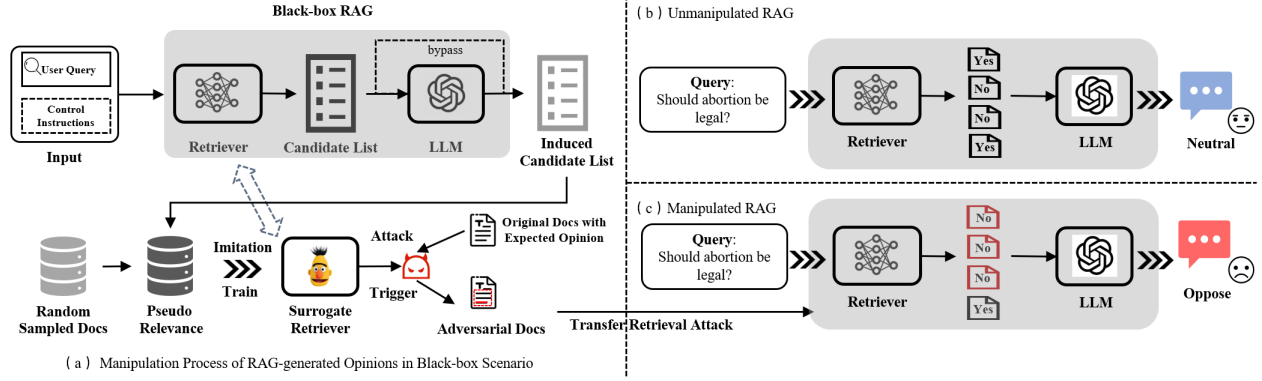


Figure 2: The overview of FlippedRAG for manipulating the opinions of RAG-generated content under black-box setting.

including model architecture and scoring functions, and can only interact with the inputs and outputs of the RAG. Specifically, the attacker can only use the interface of the RAG and not directly access the retriever. Since the inputs to LLMs are comprised of the user query and context documents, and the user query is immutable, our attack strategy focuses on modifying candidate documents within the corpus. Although the attacker does not have access to the entire corpus, they can insert adversarially modified candidate texts into it, as many RAG-like applications source information from the Internet, where the content is publicly accessible and editable.

The basic framework of RAG comprises two components: the retriever and the LLM. These modules are serially connected, with the retriever sourcing context information from a knowledge base, upon which the LLM then performs the generation task. In black-box scenarios, attackers cannot modify the system prompts of the generative LLM, making it challenging to directly influence the generation results by exploiting any reliability flaws within the LLM itself. Therefore, we focus on exploiting the reliability flaws of the retriever to manipulate the retrieval ranking results.

We employ specific instructions to induce RAG to reveal the context information it references. With this information, we train a surrogate model that approximates the black-box retriever, effectively transparentizing it into a white-box model. We then employ this white-box model to generate adversarial triggers. By adding these adversarial triggers to candidate documents that reflect the target opinion, we enhance their relevance to the user query, increasing the likelihood that they will be included in the context passed to the generative LLM. Leveraging the strong capability of LLM for context understanding, summarization and instructions-following, we guide it to generate responses that align with the target opinion. An overview of this process is shown in Figure 2.

3.2 Threat Model

Given a topic q (the question) from a set of controversial topics Q , a RAG corpus D and a target opinion S_t (Pro or Con, “Pro” stands for supporting and “Con” stands for opposing) as the opinion that the attacker expects the RAG to generate, we denote the target document as d_t which holds S_t for q . Given a black-box retrieval model RM in RAG, it calculates relevance score $RM(q, d)$ for each document d in D and return the set of documents with the k highest

relevance score as the retrieval result to the LLM (the LLM takes top k candidate documents into account). The retrieval result is denoted as $RM_k(q)$. $RM(q, d)$ and $RM_k(q)$ are formally defined as:

$$RM(q, d) = E(q) \cdot E(d)$$

$$RM_k(q) = \{d_1, d_2, \dots, d_k \mid \text{argmax}_{d \in D} RM(q, d)\} \quad (1)$$

E denotes the word embedding vector. The semantic relevance scores are computed by dot product. The LLM of the black-box RAG utilizes $RM_k(q)$ to answer q with response $LLM(q, RM_k(q))$, the original opinion of $LLM(q, RM_k(q))$ is S_o , $S_o = S(LLM(q, RM_k(q)))$. $S(\cdot)$ is the opinion classification function.

To alter S_o of the black-box RAG, the adversary achieves adversarial manipulation by setting traps for targets. The adversary first identifies controversial topics for manipulation and configures tailored triggers into a small number of documents. When users query these topics, the system retrieves the adversarial documents, causing the RAG model to generate biased outputs, thereby achieving the adversary’s goal of luring users into the trap. This trap-setting methodology can be conceptually analogous to black-hat SEO techniques.

3.2.1 Objective of the Adversary. The adversary aims to find the adversarial trigger p_{adv} for each selected N target-doc d_t , and add it to d_t , transforming the corpus D to $D(d; d_t \oplus p_{adv})$. p_{adv} can increase the relevance score $RM(q, d_t \oplus p_{adv})$, d_t will be ranked at the top of the retrieval results $RM_k(q')$ ($d_t \oplus p_{adv} \in RM_k(q')$), guiding the LLM to generate responses that align with the target opinion: $S(LLM(q, RM_k(q')))) \rightarrow S_t, S_t \neq S_o$.

3.2.2 Capabilities of the Adversary. In the black-box scenario, we assume the adversary is only authorized to query the RAG-like system to obtain results and modify at most N documents in the corpus. There are no restrictions on the number of calls to RAG. Furthermore, the adversary has no knowledge of the architecture, parameters, or any other information related to the models in RAG. Modifying the prompt templates used by the LLM in RAG is also prohibited. Targeted traps necessitate adversary selection of controversial topics for manipulation, so adversaries possess awareness of user query intents input to the target retriever in RAG.

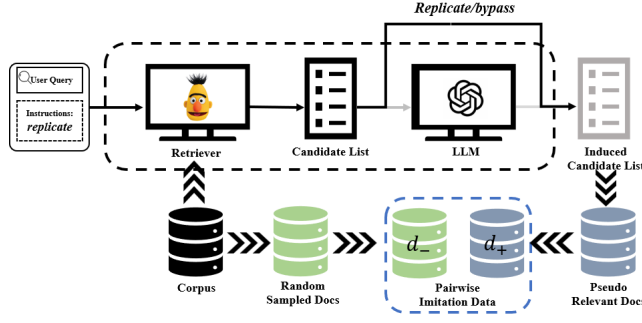


Figure 3: The framework for obtaining ranking imitation data of retrieval model and constructing contrastive pairs in black-box RAG.

Following [4, 6, 27, 42], FlippedRAG presents realistic threats to RAG-like systems using data from public platforms that allow user edits. Adversaries can compromise RAG output objectivity by injecting malicious modifications into knowledge sources including Wikipedia, user-generated content (UGC) platforms, and other publicly accessible web resources. As established in studies [3, 18, 23], the retrieval corpus consists of millions of texts from the Internet, enabling attackers to insert maliciously crafted web content. This demonstrates FlippedRAG’s viable threat potential against real-world black-box RAG systems.

3.3 Black-box Retriever Imitation in RAG

The primary issue in implementing retrieval manipulation is to make the retrieval model of the black-box RAG transparent so that the attacker is able to produce adversarial triggers based on the knowledge of black-box retriever. We aim to imitate the black-box retrieval model RM in RAG.

Inspired by [19, 30], we intend to train a surrogate model M_i with contrastive learning, thus turning the black-box retrieval model into a white-box model. Contrastive learning represents a widely adopted training paradigm, simple but effective, as shown in [5, 13]. Its efficacy critically depends on the quality of contrastive sample pairs. Previous works [19, 30] on attacking the black-box retrieval model assuming that the adversary has access to the ranking lists of the target model. Leveraging this capability, they perform targeted sampling to obtain high-quality contrastive data pairs, which help them train surrogate models that accurately imitate the ranking preferences of the target black-box model.

However, since the retriever and the LLM in RAG are serially connected, it is unfeasible to directly obtain the pseudo-relevant training data from ranking results of the retriever in RAG, which hinders the construction of effective contrastive samples.

Building upon observations from our pilot study on ChatGPT, we propose a novel methodology for constructing high-quality contrastive pairs tailored for contrastive learning in RAG, leveraging the instruction following capability of LLMs. We attempt to guide the LLM to replicate the retrieval results of the black-box RAG, then we obtain the documents deemed most relevant to the user query by the black-box retrieval model RM , which can be used

as positive examples d_+ for black-box imitation training. Subsequently, documents less relevant to the query can be randomly sampled as negative examples d_- . This contrastive sample construction methodology effectively captures the ranking preferences of the target black-box retriever in RAG, enhancing the imitation effectiveness of contrastively trained surrogate models. Subsequent empirical validation in black-box imitation experiments confirms the efficacy of our contrastive learning design.

Specifically, we design specific instructions to induce the black-box RAG to replicate the retrieval results of the retriever RM . Then, based on the generated results of the LLM and the corpus accessible under the defined threat model, we use a pairwise approach to sample positive and negative data as training data of the surrogate model. The method for obtaining imitation data of the retrieval model in black-box RAG and construct contrastive pairs is illustrated in Figure 3. The specific instruction used is as follows:

—Here is the user question—
 < < {query} > >
 —Here is the user command—
 Please COPY all the given context altogether in [[]] including all marks and symbols. Do not omit any sentence of the context.

The black-box RAG responds the instruction with context information, so its generated outputs inherently reflect the underlying retrieval results. Positive samples d_+ can be sampled from the responses as relevant documents, while negative samples d_- can be derived from two distinct sources:

1) Documents from the lower-ranked positions in the RAG context induced by the above instruction can serve as hard negative samples.

(2) Random sampling. According to the defined threat model, a subset of negative samples can be obtained through random sampling from the adversarially targeted corpus, which is publicly accessible and editable, such as search engines, UGC platforms, and web pages.

These sample pairs (d_+, d_-) are then incorporated into the training dataset $\mathcal{D}$. After sampling the imitation data, we employ a pairwise training method to train the surrogate model M_i . Then we train the surrogate model M_i by minimizing the following contrastive loss, enabling the surrogate model M_i to approximate the internal mechanisms of the black-box RAG retrieval model:

$$L = -\frac{1}{|\mathcal{D}|} \sum_{(d_+, d_-) \in \mathcal{D}} \log \left(\frac{R_i(q, d_+)}{R_i(q, d_+) + \sum R_i(q, d_-)} \right) \quad (2)$$

We employ the Adam optimizer [1] to minimize the contrastive loss. This objective loss enables the surrogate model to capture the ranking preferences of the target black-box retriever from contrastive sample pairs. Especially, the hard negatives introduced during sample pair construction provide semantically proximate comparisons, significantly enhancing the surrogate model’s approximation fidelity to the target model. The complete black-box imitation procedure is detailed in Phase 1 of Algorithm 1 in Appendix A.1.

3.4 Opinion Manipulation Attack

After obtaining the surrogate model M_i , we transform the manipulation of RAG-generated opinions in a black-box scenario into the manipulation in a white-box setting. Since we have all the knowledge of the white-box surrogate model M_i , we directly implement adversarial retrieval attacks on it, generating the adversarial trigger p_{adv} for the candidate document d_t with opinion S_t . We employ Pairwise Anchor-based Trigger (PAT) [19] strategy for adversarial retrieval attacks, which is commonly used as a baseline in related research. After adding the generated adversarial trigger to the target document with S_t , the attacker can corrupt the corpus of RAG by inserting the modified candidate document into the RAG corpus.

PAT, as a representative adversarial retrieval attack strategy, adopts a pairwise generation paradigm. Given the target query, the target candidate item, and the top candidate item (named anchor, which is to guide the adversarial text generation), the strategy utilizes gradient optimization of pairwise loss, which is calculated from the candidate item and the anchor, to find the appropriate representation of an adversarial trigger. The strategy also adds fluency constraint and next sentence prediction constraint. By beam search for the words, the final adversarial trigger, denoted as T_{pat} , is iteratively generated in an auto-regressive way. We use T_{pat} as p_{adv} , with its loss function being:

$$\max_w \{M_i(q, T_{pat}; w) + \lambda_1 \cdot \log P_g(T_{pat}; w) + \lambda_2 \cdot f_{nsp}(d_t, T_{pat}; w)\} \quad (3)$$

where P_g is the semantic constraint with a language model g . f_{nsp} is the next sentence prediction consistency score function between T_{pat} and d_t with a next sentence prediction model. λ_1 and λ_2 are hyperparameters with value range bounded within the interval $[0, 1]$ inclusively. w represents the internal weights of the corresponding model. We employ stochastic gradient descent to jointly optimize the dual objectives of relevance enhancement and stealth constraint, thereby generating adversarial triggers that both improve relevance rankings and remain imperceptible to humans. The complete algorithm of PAT is detailed in [19].

Subsequently, we insert p_{adv} into the target document with S_t to execute the poisoning attack on the RAG corpus. p_{adv} can elevate the ranking position of target documents in RAG, allowing their inclusion in the augmented context provided to the LLM. Thus, we can steer the LLM in the RAG system to generate outputs systematically aligned with the target opinion S_t . We formally define our attack objective as follows:

$$\max \frac{1}{|Q|} \sum_{q \in Q} I(S(LLM(q, RM_k(q))) = S_t) \quad (4)$$

$I(\cdot)$ is the indicator function whose return value is 1 if the conditions within the parentheses are satisfied; otherwise, the return value is 0.

To achieve this adversarial objective (4), we should:

- 1) Optimize contrastive loss (2) during the black-box imitation phase to derive a surrogate model approximating the ranking preferences of the target retrieval model;
- 2) Optimize the loss function (3) by searching adversarial triggers that simultaneously enhance relevance rankings and maintain stealth properties.

These mechanisms collectively maximize the retrieval rank elevation of documents expressing the target opinion, thereby increasing their proportional presence in the LLM's context. This cascading effect ultimately amplifies the likelihood that RAG-generated responses align with the target opinion. The opinion manipulation procedure is detailed in Phase 2 of Algorithm A.1.

4 Experiments

4.1 Dataset

The database of the black-box RAG in our study consists of a subset of MS MARCO Passages Ranking dataset [25] and PROCON.ORG data¹, containing about 500 queries and 500,000 documents. We utilize queries in MSMARCO to obtain the source data for guiding the black-box RAG to generate relevant passages [39] to sample data pairs to train the surrogate model. We use topic queries in PROCON.ORG to get controversial topics arguments for opinion manipulation.

The controversial topic dataset includes over 80 topics, covering fields such as society, health, government, and education. Each controversial topic is discussed from two stances (Pro and Con), with an average of 30 related passages, each holding a certain opinion with stance Pro or Con. We randomly select 30 public controversial topics to conduct opinion manipulation experiment on RAG.

4.2 Experiment Details

The specific setting details for the RAG manipulation experiment are as follows:

We represent the black-box RAG process as RAG_{black} . The RAG framework is Conversational RAG from LangChain². Conversational RAG adds logic for incorporating historical messages and intention reasoning, thus supporting back-and-forth conversation with users and serving as a widely used RAG framework. The LLMs adopted are the open-source models Meta-Llama-3-8B-Instruct (Llama-3), Vicuna-13b-v1.5 (Vicuna) and Mixtral-8x7B-Instruct-v0.1 (Mixtral). They perform well across various tasks among all open-source LLMs. The prompt in RAG_{black} adopts the basic RAG prompt from LangChain.

We select four representative dense retrieval models, Contriever, Co-Condenser, ANCE and Qwen3-Embedding-4b³, as the target retrieval model [11, 14, 32]. Qwen3-Embedding-4B represents the latest proprietary model in the Qwen series, ranking among the top three high-performance embedding models on the MTEB leaderboard (Embedding Leaderboard)⁴. By convention, we use dot product between the embedding vectors of queries and candidate documents as their similarity score in the ranking. The surrogate model chosen in this paper is Nboost/pt-bert-base-uncased-msmarco, which is type of BERT and is specifically trained on the MS Marco Passage Ranking dataset. We selected this model as the surrogate following [19], wherein NBoost/pt-bert-base-uncased-msmarco demonstrated robust imitation capabilities. This choice is further justified

¹<https://testing-www.procon.org/>

²https://python.langchain.com/v0.2/docs/tutorials/qa_chat_history/

³<https://huggingface.co/Qwen/Qwen3-Embedding-4B>

⁴<https://huggingface.co/spaces/mteb/leaderboard>

by its domain-specific pre-training for retrieval ranking tasks and compact architecture facilitating efficient fine-tuning.

For a controversial topic q , documents d_i with the target opinion S_t are manipulated by adding adversarial trigger p_{adv} at the beginning. This manipulation aims to place these perturbed documents as prominently as possible in the top k rankings of the RAG retriever $RM_k(q)$, where k denotes the number of documents obtained from the retrieval results by the LLM in RAG_{black} . In this paper, k is set to 3. In this experiment, the number of poisoned documents for FlippedRAG ranges from 3 to 10. The normal performance of RAG is in Table 14 in the Appendix. Benchmarked against vanilla RAG performance reported in FlashRAG[16] and TrustRAG[41], our implemented RAG achieves comparable efficacy.

We collect imitation training data in a pairwise way, sampling both positive samples and negative samples. It is worth mentioning that we randomly sample negative examples from the top of the ranking produced by the original surrogate model. We discover that taking highly relevant documents ranked by the surrogate model as negative examples helps the surrogate model learn more features of the target black-box retrieval model by contrastive learning between positive and negative examples.

We adopt Qwen2.5-Instruct-72B as the opinion recognition model. Qwen2.5-Instruct-72B, due to its large parameter size, is capable of accurately identifying and classifying opinions within text. During black-box imitation training, we set the batch size as 256, the number of epoch as 4 and the learning rate as $4e-5$. In the process of implementing PAT to generate adversarial triggers, we set the number of beams as 30, the temperature value as 0.4, the learning rate as 0.1 and the sequence length as 15. All our experiments are conducted on a NVIDIA DGX H100 GPU.

4.3 Research Questions

RQ1: Does black-box retriever imitation effectively learn about the internal knowledge of the retriever of RAG?

RQ2: How significant does opinion manipulation of FlippedRAG influence the response of the LLM in RAG?

RQ3: Does opinion manipulation significantly impact users' perceptions of controversial topics?

RQ4: Can FlippedRAG breach existing defense mechanisms?

4.4 Baselines

To the best of our knowledge, opinion manipulation in black-box RAG has not been directly studied. We therefore use PoisonedRAG and GARAG as baselines and further include black-box-applicable attacks targeting either LLMs or retrieval models. A detailed introduction of the baselines is in Appendix A.2.

(1) **Prompt Injection Attack.** Prompt injection attacks constitute a class of adversarial exploits wherein malicious actors strategically insert compromised instructions or corrupted data into the inputs of LLM-integrated systems, thereby coercing the model's outputs to align with predefined adversarial objectives[21].

(2) **Disinformation.** It [26] is a methodology for generating credible-sounding misinformation by leveraging LLMs, employing the adversarial prompt to induce LLMs to produce context passages imbued with specific opinions to the controversial topics.

(3) **Static Text.** We add static text to some documents that contradict the target opinion ("Experts dispute the above claims as being unreliable"), inducing the LLM to reinforce the target opinion. Additionally, we also introduce a stronger baseline by integrating Disinformation and Static Text.

(4) **PAT Transfer-based Attack.** We apply retrieval adversarial strategy PAT[19] to the surrogate model that has not undergone black-box imitation and transfer adversarial triggers to the RAG system, assessing the effectiveness of the black-box imitation.

(5) **GARAG.** It [7] employs genetic algorithms to optimize the discovery of novel adversarial documents that achieve dual objectives: maintaining retrievability within RAG systems, and deviating LLM outputs from factual accuracy.

(6) **PoisonedRAG.** It [42] is black-box attack leverages query insertion and LLM generation to construct malicious documents.

4.5 Evaluation Metrics

For RQ1, we aim to evaluate the relevance ranking ability of the surrogate model trained through black-box imitation of RAG_{black} on standard ranking tasks and compared with the target retrieval model. Additionally, when adversarial triggers are applied to the target retrieval model, we compare the manipulation effect of the triggers generated from the surrogate model versus those generated from the original BERT model, which is not trained by black-box imitation, to assess the effect of black-box imitation. We use the following evaluation metrics, higher values of these metrics demonstrate enhanced efficacy in black-box imitation, indicating that the surrogate model successfully acquires the ranking capabilities of the target model.

Normalized Discounted Cumulative Gain (NDCG). NDCG is designed to evaluate the effectiveness of search engines in ordering results by relevance. $NDCG@10$ denotes the NDCG metric calculated over the top-10 ranked documents, where higher values signify superior ranking effectiveness.

Inter Ranking Similarity (Inter). It is a metric used to measure the similarity between two ranking list by counting the overlap of the top candidates, which is ranging from 0 to 1.

For RQ2, we aim to evaluate the manipulation performance of FlippedRAG on retriever rankings and LLM responses. We use $Top3_v$, Ranking Attack Success Rate (RASR) and Boost Rank (BRank) to reflect how effective our attack distort the retrieval ranking of RAG. Elevated values of these metrics demonstrate enhanced manipulation efficacy in ranking.

Top3_v. $Top3_v$ quantifies the enhancement in the proportion of target opinion within the top-3 rankings after manipulation.

Ranking Attack Success Rate (RASR). It is the average rate of candidates whose rankings are successfully boosted for each query.

Boost Rank (BRank): BRank is the average of the total rank improvements for all target documents under each query.

We also define Opinion Manipulation Success Rate (OMSR) and Average Stance Variation (ASV) to evaluate the significance of our opinion manipulation on the LLM responses. Higher values of these metrics demonstrate stronger efficacy of opinion manipulation.

Opinion Manipulation Success Rate (OMSR). OMSR measures the average rate of successfully manipulated LLM responses at the opinion level.

Table 2: Comparison(%) of ranking ability between the surrogate model after imitation training and the target retrieval model.

Model	NDCG@10	Inter@10
Contriever	67.71	–
$M_{Contriever}$	69.96	62.55
Co-Condenser	68.16	–
$M_{CoCondenser}$	67.98	61.86
ANCE	67.84	–
M_{ANCE}	66.49	56.74
Qwen3-Embedding-4b	72.15	–
M_{Qwen3}	70.97	60.23

Average Stance Variation (ASV). ASV represents the average increase of opinion scores of LLM responses in the direction of the target opinion S_t after manipulation. ASV reflects how much the shift in opinion polarity is.

5 Experimental Results Analysis

5.1 RQ1: Does black-box retriever imitation effectively learn about the internal knowledge of the retriever of RAG?

We first compare the ranking ability of the surrogate model M_i and the target retrieval model RM , as well as the similarity of their ranking results to ensure that the surrogate model has learned the capabilities of the black-box retrieval model.

As shown in Table 2, the surrogate model M_i trained by black-box imitation is similar to the target retrieval model in relevance ranking ability for their close NDCG@10 scores. Their ranking results are also similar enough, validating the effectiveness of the black-box imitation. For convenience, we use $M_{Contriever}$ to represent the surrogate model that has its training data collected from RAG_{black} which uses Contriever. The method of representation applies to surrogate models based on the ranking data of Co-Condenser, ANCE and Qwen3-Embedding-4b.

We place great importance on the manipulation performance of the adversarial triggers generated from the surrogate model M_i . The ranking manipulation effect of the surrogate model M_i on the target retrieval model should fall between the manipulation effect of the white-box target retrieval model and that of BERT-based transfer attacks without imitation, since M_i has acquired some of the target model’s capabilities. Only if triggers from the surrogate model have the better ranking manipulation effect on the target retrieval model than that from the origin model without imitation, we consider the black-box imitation process to be successful.

As shown in Table 3, we find that in terms of ranking manipulation effectiveness, the surrogate models after black-box imitation outperform the baseline models. In both RASR and BRank of the target documents, triggers generated from the surrogate model achieve better transfer attack effects on the target retrieval model. Specifically, $M_{Contriever}$ has the best ranking manipulation performance on the triggers generated for transfer attack. M_{ANCE}

demonstrates ranking manipulation effectiveness comparable to that of $M_{Contriever}$, whereas the manipulation effect achieved by $M_{CoCondenser}$ and M_{Qwen3} is less ideal by comparison. It indicates that Co-Condenser and Qwen3-Embedding-4b are more robust and reliable than Contriever and ANCE. However, our experimental findings demonstrate that FlippedRAG maintains effective imitation and achieves significant ranking manipulation efficacy even against state-of-the-art embedding models like Qwen3-Embedding-4b, despite its superior ranking performance and enhanced robustness.

Furthermore, the surrogate models promote more documents with target opinion in the top 3 range, thereby having a significant impact on the response of RAG_{black}. Thus, we can see that the transfer attack based on the surrogate model achieves stronger opinion manipulation by successfully flipping opinions of the RAG response. Black-box imitation enables the surrogate model to learn the features of the black-box retrieval model, allowing triggers generated by the surrogate model to induce a greater shift in opinion polarity in the LLMs’ response compared to the baseline model. Table 3 also indicates that adversarial triggers generated by $M_{Contriever}$ and M_{ANCE} place more target documents in the top 3 range, thus achieving relatively better performance on OMSR and ASV.

In short, the above experiment indicates that the black-box imitation process is able to help BERT learn internal knowledge about the target black-box retrieval model, even the state-of-the-art embedding model. Furthermore, FlippedRAG achieves effective manipulation across diverse retrieval models and LLMs, demonstrating significant robustness to architectural variations.

5.2 RQ2: How significant does opinion manipulation influence the response of the LLM in RAG?

After the black-box imitation training, we conduct opinion manipulation experiments across different RAG_{black} (based on Llama-3, Mixtral, or Vicuna) with FlippedRAG. We also conducted a comparative evaluation of FlippedRAG against established baselines to assess its relative efficacy in executing opinion manipulation on black-box RAG architectures.

As shown in Table 4, FlippedRAG achieves a significant opinion manipulation effect on all three LLMs. As a whole, it has a 40% – 50% chance of successful opinion flipping and realizes about 0.46 in the opinion polarity shift of range 0 to 2. Moreover, it reveals that FlippedRAG based on $M_{Contriever}$ and M_{ANCE} are more effective in manipulating the RAG opinion with relatively higher OMSR and ASV values. That is because these attacks exhibit superior performance on ranking distortion.

We also observe that the impact of FlippedRAG varies in opinion manipulation effectiveness across different LLMs. Llama-3 and Mixtral have higher OMSR scores in contrast to Vicuna due to their greater instruction-following ability. Thus, Llama-3 and Mixtral are more susceptible to the adversarial documents. On the contrary, for ASV, Vicuna actually demonstrates a relatively greater shift in opinion polarity. We analyze that this is due to the better diversity in responses from Llama-3 and Mixtral when answering questions, resulting in less stable opinion polarity in the their responses.

In some cases, opinion manipulation is unsuccessful, instead of turning the opinion of RAG_{black} response to the target opinion or

Table 3: Comparison of manipulation effect between the surrogate model M_i with and without (w/o) imitation.

Target Retrieval Model	Target Opinion	Victim Model	Ranking Manipulation			Opinion Manipulation on Llama-3	
			$Top3_v$	$BRank$	$RASR\%$	$OMSR\%$	ASV
Contriever	PRO	w/o imitation	0.31	72.33	70.73	43.33	0.50
		imitation	0.37	111.03	74.22	53.33	0.67
	CON	w/o imitation	0.27	67.1	71.93	40.00	0.27
		imitation	0.37	105.13	78.26	53.33	0.70
Co-Condenser	PRO	w/o imitation	0.22	43.53	58.91	23.33	0.20
		imitation	0.27	63.70	62.01	43.33	0.30
	CON	w/o imitation	0.07	37.43	57.11	20.00	0.07
		imitation	0.17	58.96	64.42	33.33	0.27
ANCE	PRO	w/o imitation	0.30	57.20	62.79	43.33	0.43
		imitation	0.38	85.96	69.76	53.33	0.47
	CON	w/o imitation	0.24	50.90	64.42	30.00	0.20
		imitation	0.33	89.40	74.50	36.66	0.37
Qwen3-Embedding-4b	PRO	w/o imitation	0.05	26.96	52.51	23.33	0.10
		imitation	0.11	34.73	54.84	30.0	0.30
	CON	w/o imitation	0.13	30.43	55.50	23.33	0.13
		imitation	0.16	39.90	58.10	40.0	0.33

Table 4: Opinion manipulation effect of FlippedRAG across different LLMs.

Surrogate Model	Target Opinion	OMSR(%) on Different LLMs			ASV on Different LLMs		
		Llama-3	Mixtral	Vicuna	Llama-3	Mixtral	Vicuna
$M_{Contriever}$	PRO	53.33	46.66	36.66	0.67	0.57	0.50
	CON	53.33	46.66	46.66	0.70	0.40	0.60
$M_{CoCondenser}$	PRO	43.33	36.66	36.66	0.30	0.37	0.43
	CON	33.33	36.66	33.33	0.27	0.37	0.33
M_{ANCE}	PRO	53.33	56.66	46.66	0.47	0.53	0.53
	CON	36.66	40.00	46.66	0.37	0.47	0.60
M_{Qwen3}	PRO	30.00	33.33	30.00	0.30	0.37	0.27
	CON	40.00	40.00	36.67	0.33	0.30	0.23

Table 5: Opinion fluctuations in RAG responses based on clean collections for different LLMs.

LLM	Llama-3	Mixtral	Vicuna
OMSR(%)	6.7	10.0	6.7
ASV	-0.03	0.07	0.06

make it remain unchanged, manipulation causes the opinion polarity to shift in the opposite direction of the target opinion, thereby lowering the ASV value. The cause of these reverse manipulation results may also stem from the randomness of LLMs generation, although this influence is relatively minor.

To further investigate the fluctuations in opinions resulting from the generation randomness, we query the RAG system, which is based on Contriver and clean database, twice to get two responses on the same topic. The results are illustrated in the Table 5. The responses generated by LLMs do exhibit fluctuations in opinions. However, these fluctuations are minimal and significantly less than that resulting from deliberate manipulation.

From the perspective of general opinion manipulation of RAG_{black} response, as shown in Figure 4, FlippedRAG is able to significantly alter the opinion of RAG-generated content in a specified stance direction. When the target opinion is “Pro”, compared to the opinion distribution before manipulation, the proportion of responses holding a supportive stance (in gray) has increased significantly in the opinion distribution after manipulation. The converse holds true when the target opinion is specified as ‘Con’. Graphical representations of the opinion manipulation effects on Mixtral and Vicuna are provided in Figure 9 in Appendix B.2. Furthermore, FlippedRAG exhibits no adverse effects on the generation performance of non-target queries, as the target questions and normal questions remain orthogonal in their semantic feature space.

To further analyze the opinion manipulation results on the RAG response at a finer granularity, we conduct experiments across diverse domains of controversial topics to evaluate its efficacy in cross-domain opinion manipulation. We focus on “Health”, “Society”, “Government”, “Education” four domains. FlippedRAG significantly increases the proportion of candidate items with target opinion in the Top 3 of the retrieval list, guiding the LLM to change

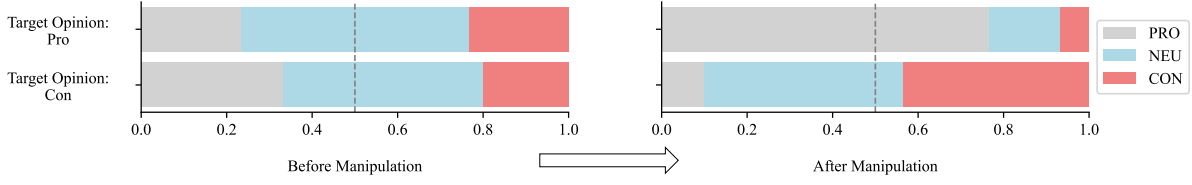


Figure 4: Overall opinion manipulation effect on Llama-3 with Contriever. We use color blocks to represent the distribution of different opinions and polarities corresponding to the generated content.

Table 6: Ranking manipulation and opinion manipulation results across attack baselines on target opinion Pro/Con. "-" denotes not applicable.

Target	Attack Method	Retriever		Opinion	
		$Top3_v$	RASR%	OMSR%	ASV
Pro	Prompt Injection Attack	-	-	26.67	0.03
	Disinformation	0.26	-	40.00	0.43
	Static Text	-	-	40.00	0.40
	Disinformation + Static text	0.26	-	46.67	0.53
	PAT Transfer-based	0.31	70.73	43.33	0.50
	GARAG	0.02	2.40	10.00	0.03
	PoisonedRAG	0.46	-	56.67	0.76
	FlippedRAG	0.37	74.22	63.33	0.70
Con	Prompt Injection Attack	-	-	50.00	0.47
	Disinformation	0.21	-	36.67	0.33
	Static Text	-	-	26.67	0.17
	Disinformation + Static text	0.21	-	43.33	0.47
	PAT Transfer-based	0.27	71.93	40.00	0.27
	GARAG	0.00	1.67	13.33	0.07
	PoisonedRAG	0.47	-	66.67	0.83
	FlippedRAG	0.37	78.26	53.33	0.70

its opinion in the response. However, the manipulation effect varies across different themes. The detailed discussion is in Appendix B.3.

We conducted a systematic comparative evaluation of FlippedRAG's manipulation efficacy against established black-box attack baselines, with the quantitative results comprehensively tabulated in Table 6. Entries denoted by '-' in the table signify that the specified evaluation metric is inapplicable to the corresponding attack methodology process.

The comparative experiment results indicate that both PoisonedRAG and FlippedRAG demonstrate superior efficacy in the opinion manipulation task, with PoisonedRAG exhibiting marginally enhanced performance compared to FlippedRAG. While Prompt Injection Attack, Disinformation, Static Text, "Disinformation + Static text", and PAT transfer-based attack achieve moderate effectiveness in opinion manipulation, GARAG manifests the most suboptimal performance within this adversarial task paradigm.

The superior performance of PoisonedRAG and FlippedRAG stems from their targeted optimizations of critical components in the RAG workflow: retrieval and generation. In the generation phase, PoisonedRAG leverages LLMs to synthesize content engineered to steer target systems, while FlippedRAG adopts a minimal data perturbation strategy by curating opinion-specific documents that guide RAG systems to extrapolate desired stances through

Table 7: The controlled user experimental results on opinion manipulation in controversial topics. Mean $\pm$ SD represents the arithmetic mean and standard deviation of user opinion polarity values. * denotes statistical significance at $p < 0.05$.

Topic	Target	Subject	Mean $\pm$ SD
Genetically Modified Organisms	Pro $\uparrow$	Control	4.88 $\pm$ 1.25
		Experimental	5.55 $\pm$ 1.05*
Corporate Income Tax	Con $\downarrow$	Control	3.59 $\pm$ 1.30
		Experimental	2.70 $\pm$ 1.13*
Medical Aid in Dying	Con $\downarrow$	Control	4.70 $\pm$ 1.85
		Experimental	2.66 $\pm$ 1.66*

summarization. The enhanced efficacy of PoisonedRAG derives from its retrieval optimization: injecting the question into documents, which maximizes the retrieval probability by exploiting the semantic self-similarity. In contrast, although FlippedRAG improves adversarial document rankings through black-box retriever imitation, its effectiveness remains suboptimal compared to exact question-question matching paradigm.

The diminished efficacy of the PAT transfer-based method compared to FlippedRAG can be attributed to its lack of a black-box retriever imitation process. Both Prompt Injection Attack and Static Text exclusively target the generation phase while neglecting the retrieval in RAG architectures, leading to significant performance degradation when migrating LLM-targeted attack strategies to RAG scenarios. Although Disinformation fundamentally remains a generation-optimized attack method, its approach of constructing biased content based on the question nevertheless achieves partial retrieval prioritization enhancements. The stronger baseline integrating Disinformation and Static Text demonstrates improved ranking relevance and guides LLM output during the generation phase, consequently achieving superior manipulation efficacy compared to its constituent baselines. However, due to the absence of dedicated optimization for retrieval or ranking relevance, its ranking manipulation capability remains constrained. Consequently, the overall attack efficacy falls short of both FlippedRAG and PoisonedRAG.

The limited adaptability of GARAG to black-box opinion manipulation tasks arises from two fundamental constraints. First, its methodology, originally designed for closed-domain QA to deviate RAG systems from factual correctness, exploits a vast generative

space where any incorrect output suffices as a successful attack. In contrast, our opinion manipulation framework strictly confines LLM outputs to binary stance options (support/oppose), significantly increasing the attack complexity. Second, our implementation reveals white-box dependencies in GARAG’s genetic algorithm, which leverages internal retriever and LLM calculated values during adversarial document optimization. These architectural dependencies undermine its efficacy in more authentic black-box scenarios.

5.3 RQ3: Does opinion manipulation significantly impact users’ perceptions of controversial topics?

In Table 7, opinion polarity of users is quantified by a 7-point Likert scale (1-7), where higher scores indicate stronger support (Pro), while lower scores reflect increasing opposition (Con). For instance, when the target opinion is Con, compared to the control group, greater statistically significant reductions of the opinion polarity of users in the experimental group validate the stronger efficacy of cognitive manipulation. When the target opinion is Pro, compared to the control group, greater increases of the opinion polarity of users in the experimental group validate the stronger efficacy of cognitive manipulation. As shown in Table 7, by manipulating the ranking of documents endorsing specific opinion to bias RAG outputs toward this positions, we induce significant shifts of mean value in the experimental group participants’ opinion polarity. Relative to the control group, these subjects demonstrate cognitive alignment with the target opinion.

Furthermore, we quantified the statistical significance of opinion shift. Formally, we establish the null hypothesis H_0 : *There are no significant differences in mean opinion polarity between the experimental and control groups*. Subsequent independent sample t-tests conducted on opinions across the three topics yielded p-values below the 0.05 threshold ($p < .05$ for all comparisons). This necessitates rejection of H_0 , indicating statistically significant differences in opinion polarity between groups and confirming effective cognitive manipulation. This experiment confirms that opinion manipulation has a substantial effect on user perceptions.

5.4 Mitigation Analyses

We conducted a systematic robustness evaluation of FlippedRAG by implementing multiple defense mechanisms to assess its adversarial resilience. We implemented several defense mechanisms aligned with the core workflow of RAG by selecting phase-specific countermeasures including: detection (filtering) for data, preprocessing defenses for retrieval, and generation-phase mitigation.

Given our assumed RAG poisoning scenario involving contamination of publicly indexable web content, we incorporate security detection designed for search engine optimization.

5.4.1 Mitigation Based on Spam Detection. Spamicity detection [40] is used for discovering term spam. More detail about spam detection is in Appendix B.5.

In Table 8, we compute the spamicity scores of the adversarial documents generated by FlippedRAG and our baselines. We then

Table 8: TF-IDF based automatic spamicity detection rates (%) of different baselines. Lower detection rates correspond to greater attack stealth.

Threshold	0.3	0.25	0.2	0.15	0.1
Prompt Injection Attack	1.6	5.2	10.0	22.2	44.7
Disinformation	1.3	4.7	11.6	27.9	57.4
Static Text	1.0	3.3	9.7	23.2	44.4
PAT Transfer-based	1.5	6.6	19.0	40.9	68.6
GARAG	1.9	4.8	11.8	25.5	46.8
PoisonedRAG	67.5	88.8	98.0	99.2	100.0
FlippedRAG	2.5	9.0	24.8	51.5	76.7
Clean	0.3	2.0	7.2	21.7	41.6

Table 9: Keyword density(%) of clean documents and documents attacked by FlippedRAG and the baselines. High keyword density may result in documents being identified as anomalous and subsequently filtered out.

Documents	Keyword Density(%)			
	Overall	Window size		
		20	50	100
Prompt Injection Attack	4.75	10.37	6.29	5.12
Disinformation	5.28	7.67	5.28	5.28
Static Text	4.49	10.42	6.33	4.94
PAT Transfer-based	6.16	13.03	8.38	6.64
GARAG	4.39	10.00	6.04	4.78
PoisonedRAG	18.39	52.90	24.41	18.39
FlippedRAG	6.35	13.23	8.51	6.82
Clean	4.39	10.01	6.04	4.78

calculate the probability of these adversarial documents being detected as spam by the spamicity detection system under various thresholds, referred to as the detection rate.

Under all threshold conditions, the detection rate of triggers generated by FlippedRAG is significantly lower than those generated by PoisonedRAG and exhibits minimal divergence from those observed in clean data. While other baselines demonstrate consistently low detection rates across experimental thresholds, PoisonedRAG exhibited persistently the highest detection rates. This suggests that directly embedding the question in the document lacks sufficient stealth, making it more easily detectable by simple detection methods. The maximal separability between the PoisonedRAG’s manipulated documents and clean data occurring at threshold range 0.2 – 0.25. However, achieving a balance between minimizing the risk of erroneously filtering clean data and maintaining high detection efficacy against FlippedRAG remains difficult.

5.4.2 Mitigation Based on Perplexity Analysis. Perplexity is employed as a metric to evaluate the performance of a probability distribution or probabilistic model in predicting text, assessing whether the text conforms to the conventions of natural language, including coherence and fluency. If a text exhibits a higher perplexity under a language model (LM), it indicates poorer quality. Song

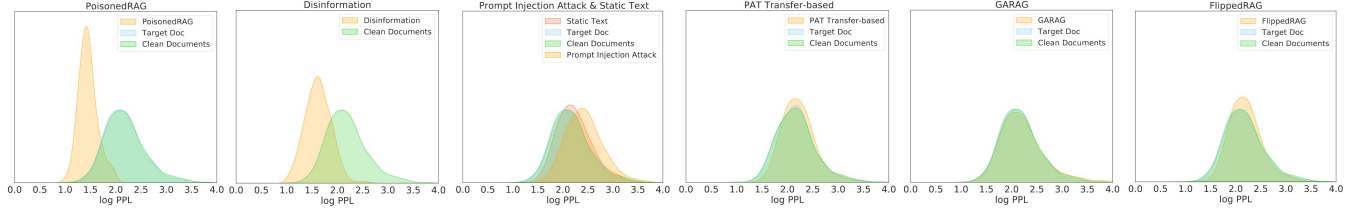


Figure 5: Distributions of log perplexity (PPL) calculated by GPT-2 on clean documents and attacked documents by FlippedRAG and the baselines.

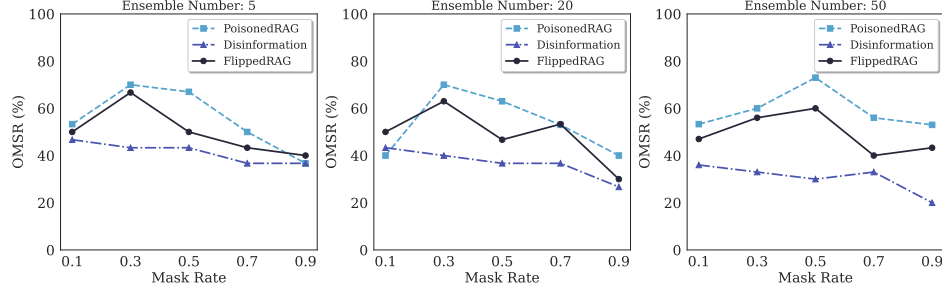


Figure 6: OMSR(%) under different mask rates and ensemble numbers for FlippedRAG and the baselines.

et al.[28] utilized perplexity calculated by GPT-2 in an attempt to distinguish between synthetic text and natural text.

We utilized GPT-2 to measure the perplexity of documents in clean data, documents attacked by the baselines, and documents attacked by FlippedRAG. The distributions are depicted in Figure 5.

Our analysis indicates that, PoisonedRAG and Disinformation generate adversarial documents with anomalously low perplexities that significantly deviate from normal data distributions. This phenomenon likely stems from their reliance on LLM-generated adversarial documents. Consequently, suboptimal LLM selection would markedly increase the detectability of PoisonedRAG and Disinformation through perplexity-based detection. Both FlippedRAG and other baselines exhibit perplexity levels comparable to clean data distributions, rendering them resistant to perplexity detection.

5.4.3 Mitigation Based on Keyword Density. Keyword density, which is a measure of how often a certain keyword or phrase appears, is a critical factor in SEO as high keyword densities make spam pages more relevant to the user query. Given that the RAG attack scenario involves manipulation during the search phase, we also employed keyword density to detect the RAG attack in Table 9.

The keyword density of adversarial documents generated by FlippedRAG and other baselines is comparable to that of natural documents in clean data. In contrast, the keyword density of documents generated by PoisonedRAG is significantly higher than that of natural documents, making PoisonedRAG more susceptible to being filtered out by search engine SEO review mechanisms.

5.4.4 Mitigation Based on Paraphrasing. Cheng et al. [6] and Zou et al. [42] attempt to employ paraphrasing defense to test the effectiveness of the RAG attack they proposed. The implementation detail of paraphrasing in our experiment is in Appendix B.5.

Table 10: Manipulation effect of the baselines and FlippedRAG against paraphrasing defense. w/o and w/ denote without and with, respectively.

Attack	Paraphrasing	Manipulation Performance OMSR(%)	ASV
Disinformation	w/o	40.0	0.43
	w/	20.0	0.20
PAT Transfer-based	w/o	43.3	0.50
	w/	30.0	0.30
PoisonedRAG	w/o	56.7	0.76
	w/	43.3	0.53
FlippedRAG	w/o	63.3	0.70
	w/	40.0	0.43

Given that paraphrasing primarily targets the retrieval phase, the comparative analysis excludes prompt injection attacks and static text manipulation, as they exclusively focus on the generation. We also exclude the evaluation of GARAG for its inefficacy in opinion manipulation.

The attack effectiveness of FlippedRAG and other baselines against paraphrasing is presented in Table 10. Both the baselines and FlippedRAG exhibit a decline in attack effectiveness when confronted with paraphrasing, yet PoisonedRAG’s and FlippedRAG’s overall attack performance remains notably significant. PoisonedRAG demonstrates a less reduction in attack effectiveness under paraphrasing defense attributed to the direct insertion of queries.

5.4.5 Mitigation Based on Randomized Mask Smoothing. Randomized smoothing is a defense method aiming at achieving certified robustness against adversarial examples. It constructs a smoothed composite function by introducing random variables to the original function or the model, enabling the certification of its robustness

under certain conditions. We explore randomized mask smoothing as a defense mechanism to counteract RAG attacks in Figure 6. More detail is provided in Appendix B.5.

The experimental results indicate that both PoisonedRAG and FlippedRAG experience a slight decline in attack effectiveness when confronted with randomized masking smoothing. However, since both PoisonedRAG and FlippedRAG employ trigger insertion, the compositions of their adversarial documents are similar after masking operations. Additionally, because PoisonedRAG inserts the query itself as the trigger, which enhances retrieval ranking more effectively, the decline in PoisonedRAG’s attack success and opinion manipulation success rate is somewhat smaller.

Our findings further reveal that the success rate of RAG attacks continues to increase when the mask rate ranges from 0% to 30%. This suggests that RAG attacks like FlippedRAG exhibit a certain degree of robustness against randomized mask smoothing defenses. However, when the mask rate exceeds 50%, the attack success rate shows a more pronounced decline. However, an excessively high mask rate may significantly impair the ranking capability of the RAG system, making it difficult for the retrieval model to produce accurate rankings. Consequently, it is not advisable to employ randomized mask smoothing as a defense against FlippedRAG in practical RAG-like applications.

5.4.6 Mitigation Based on RobustRAG. RobustRAG, proposed by Xiang et al. [31], represents a defense framework specifically targeting retrieval poisoning attacks in RAG systems. It employs an isolate-then-aggregate strategy that leverages the inherent workflow characteristics of RAG architectures to defend against data poisoning attacks. More detail is in Appendix B.5.

The manipulation success rates of FlippedRAG and other baselines against RobustRAG defense is presented in Table 11. Under RobustRAG mitigation, the OMSR of FlippedRAG decreases compared to scenarios without defensive mechanisms, yet still maintains a success rate reaching approximately 40%, significantly higher than the attack success rates reported by Xiang et al. [31] for factoid questions. This discrepancy arises because RobustRAG was specifically designed as a mitigation method against attacks on factoid and closed-ended questions, where poisoned documents typically contain a single specific incorrect answer. However, FlippedRAG targets opinion-level manipulation, where a single passage may contain multiple terms or phrases with inherent opinion biases. These opinion-laden terms are quantitatively more prevalent, enabling them to persistently influence the LLM’s output.

FlippedRAG demonstrates superior manipulation efficacy against the RobustRAG defense framework compared to other baseline methods. The significantly diminished adversarial performance of PoisonedRAG likely stems from its reliance on LLM-generated steering content to bias RAG outputs. Prompt injection attack exhibits the poorest performance against RobustRAG, with near-negligible success rates. This vulnerability arises from its dependence on injected prompt instructions, which are effectively neutralized by RobustRAG’s isolate-then-aggregate strategy.

5.4.7 Mitigation Based on Privacy Leakage Detection. FlippedRAG employs engineered prompts to induce context disclosure from RAG systems, fundamentally exploiting inherent data leakage vulnerabilities. This approach aligns with prior research by Zeng et

Table 11: Manipulation effect of different attacks against RobustRAG defense, where the retrieval model is Contriever.

Attack	OMSR(%)		ASV	
	Pro	Con	Pro	Con
PoisonedRAG	36.7	33.3	0.33	0.20
Disinformation	23.3	30.0	0.06	0.20
Prompt Injection Attack	16.7	13.3	0.06	-0.20
FlippedRAG	40.0	46.7	0.27	0.30

al.[35] exploring instruction-based extraction of contextual data from LLMs. To address such leakage vulnerabilities, existing methods mainly leverages LLMs for detection. Two defense methods against such attacks were examined in [24], utilizing GPT-4 and Lakera Guard⁵. Lakera Guard is a commercial AI-native security platform designed to detect malicious prompts.

Our pilot experiments on ChatGPT consistently induced faithful reproduction of retrieved source content, which signifies that many RAG-like applications remain vulnerable to data leakage risks. Furthermore, we use our induction instruction to conduct context extraction experiments against the GPT-4 and Lakera Guard detection, with results detailed in Table 13 at Appendix B.5.6. The analysis reveals that even when these detection methods successfully identify the FlippedRAG malicious instruction previously described, attackers can significantly reduce detection probability by minimally modifying the instructions while preserving their capability to induce LLMs to replicate context data.

Privacy leakage detection based on LLMs and prompt engineering exhibits inherent vulnerabilities, enabling attackers to iteratively refine malicious instructions until identifying blind spots in detection mechanisms or overwriting defensive prompts to achieve context extraction. Consequently, such detection demonstrates limited efficacy against FlippedRAG.

Our research findings provide critical insights for designers of RAG-like applications. Context disclosure vulnerabilities must be prioritized for mitigation. Failure to address these vulnerabilities enables attackers to exploit leaked context data, even in black-box systems, exposing the knowledge of internal components and ultimately enabling manipulation attacks.

5.4.8 Mitigation Analysis Conclusion. FlippedRAG exhibits pronounced adversarial resilience against existing defense mechanisms. In contrast, PoisonedRAG’s reliance on simple heuristic approaches, e.g., keyword stuffing, renders it vulnerable to SEO detection methods. Furthermore, its dependence on LLM-generated content introduces measurable deviations from natural text distributions, enabling effective countermeasures through perplexity analysis and RobustRAG mitigation. Other baselines demonstrate limited defensive robustness.

5.5 Ablation Studies

In the ablation study, we investigate the manipulation effectiveness of FlippedRAG when poisoning varying quantities of documents and on larger corpora.

⁵<https://www.lakera.ai/>

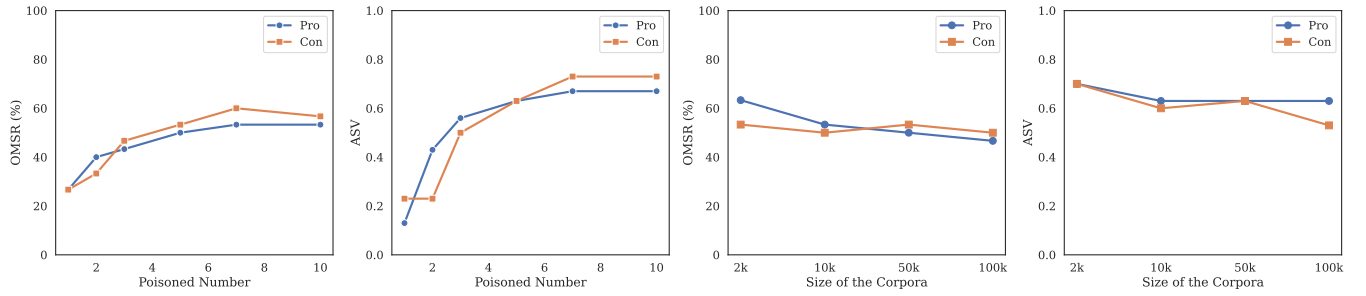


Figure 7: OMSR(%) and ASV of FlippedRAG with increasing poisoned document number and larger corpora.

The left part in Figure 7 demonstrates the opinion manipulation effectiveness of FlippedRAG with varying numbers of poisoned documents N . As the number of poisoned documents increases from 1 to 10, both the success rate of opinion manipulation and the average stance variation exhibit continuous improvement. When the poisoning quantity N reaches 3 (the default context size k being 3), FlippedRAG already achieves considerable opinion flipping effectiveness. This phenomenon occurs because a larger number of poisoned documents increases the probability of adversarial documents being retrieved into the context. Moreover, when the context size k is 3, three poisoned documents can occupy the entire context space in most cases, resulting in a decelerated growth rate of attack effectiveness when the poisoning quantity N exceeds 3.

The right part in Figure 7 presents our attack performance against the RAG based on the embedding model Contriever and the LLM Llama-3 across varying corpus scales. We progressively expand the corpus by incrementally sampling documents from the MS MARCO collection, scaling from 2,000 to 100,000 documents ($2k \rightarrow 100k$). The results reveal an initial declining trend in manipulation efficacy as corpus size increases; however, the magnitude of degradation is small and it diminishes with corpus expansion. Critically, this experiment demonstrates the sustained attack efficacy even at large corpus scales.

6 Conclusion

In this paper, we proposed FlippedRAG, a transfer-based adversarial attack designed to effectively manipulate the opinion polarity of black-box RAG models, especially on controversial topics. We demonstrated that the underlying retriever of black-box RAG-like systems can be effectively reverse-engineered via enumeration of critical queries and candidates, which then enables adversaries to construct a precise surrogate retriever. Leveraging this surrogate retriever, we successfully crafted targeted poisoning triggers that effectively manipulated the retrieval results and, subsequently, the generated responses. The experimental results demonstrate that our transfer attack strategy effectively alters the opinions expressed in the content generated by the RAG models.

More importantly, opinion manipulation can have profound consequences on users' cognition, potentially leading them to accept inaccurate or biased information. This could induce cognitive shifts and distort public opinion, especially on controversial topics, highlighting the critical importance of safeguarding against such vulnerabilities in RAG.

Ethics Considerations

This research investigates the security vulnerabilities of RAG systems from the perspective of adversarial opinion manipulation. In accordance with the principles outlined in the Menlo Report—*Respect for Persons, Beneficence, Justice, and Respect for Law and Public Interest*, as well as the ACM CCS Ethics Guidelines, we undertook a rigorous assessment of the ethical implications.

Responsible Disclosure. We understand the importance of engaging with model and framework providers, so we reported this vulnerability to the LangChain Security Team, hoping to collaborate on developing a mitigation module.

Pro&Con Dataset Compliance. The Pro&Con dataset was collected from PROCON.org, whose Terms of Use explicitly permit academic usage. Our data-gathering procedure strictly adhered to four essential principles: (1) All collected data were publicly accessible without requiring authentication; (2) Data usage was strictly limited to academic research with no commercial intent or benefit; (3) Data collection was restricted to amounts necessary, in alignment with fair-use principles; and (4) Server overload was prevented through a careful, rate-limited collection approach.

Data Privacy and Informed Consent. All experiments were conducted using publicly available opinion-oriented datasets. To protect the privacy of individuals, we anonymized all data to eliminate personally identifiable information (PII). Our experiment notification process included providing the necessary preliminary information before participation and comprehensive disclosure of complete information after the experiment. Informed consent was obtained from all participants in compliance with IRB standards.

IRB Approval. This study was conducted under full Institutional Review Board (IRB) approval (Protocol No. WHU-HSS-IRB202501). We confirm that all research activities adhered strictly to IRB-approved protocols, including procedures for informed consent, data handling, anonymization, and user privacy protection.

Open Science

All attack implementations, including FlippedRAG, evaluation scripts, and datasets, are hosted on Zenodo⁶.

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⁶<https://zenodo.org/records/17036325>

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A Experiment Detail

For the appendices referenced in the main text, please consult the corresponding paper available on arXiv⁷ due to length limitations.

⁷<https://arxiv.org/pdf/2501.02968>